

ECEN 665

Section – 601

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Lab 02: (Post-lab) report

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Texas AM University

Electrical Engineering Department

ECEN 665

Laboratory #2: Design and Understanding of RF Components: Inductors & Varactors

Objectives: To learn the fundamentals of inductors and varactors for RF CMOS circuits, to learn the design and characterization of inductors and varactors and to understand the use of ASITIC for the simulation of inductors implemented in silicon technologies.

1. VARACTORS

1.1 Explain different types of varactors that are employed in RF CMOS design, and discuss the pros and cons of each. What is the most common use of varactors in RFIC design? Evaluate each different type of varactor and their pros and cons with respect to this common application.

Introduction:

Varactors are variable capacitors whose capacitance can be controlled by a voltage. In RF CMOS design, they are crucial for numerous functions, primarily for tuning circuits. The main types of varactors used in RFIC (Radio Frequency Integrated Circuit) design are the P-N junction varactor, MOS varactor, and accumulation varactor.

Types of Varactors:

1. P-N Junction Varactor (Diode Varactor):

- *Structure:* Uses a reverse-biased P-N junction; capacitance varies with reverse voltage.
- *Pros:* Simple structure; High Q (quality factor), hence low losses; Well-understood behavior.
- *Cons:* Not easily integrated in standard CMOS without process modification; Limited tuning range; May require extra fabrication steps.

2. MOS Varactor:

- *Structure:* Utilizes a MOS transistor, typically nMOS or pMOS, configured as a capacitor with voltage-controlled capacitance.
- *Pros:* Easy to implement in standard CMOS processes; Large tuning range; Can be integrated with digital circuits.
- *Cons:* Lower Q compared to P-N junction; Capacitance nonlinearity and sensitivity to process variations; May contribute to increased noise.

3. Accumulation Varactor:

- *Structure:* A MOS varactor operated in accumulation mode, often using lightly doped regions to enhance capacitance tuning.
- *Pros:* Higher Q than conventional MOS varactors; Better linearity and tuning range; Compatible with standard CMOS, no special processing needed.
- *Cons:* Still not as high-Q as discrete P-N junctions; Performance depends on layout and oxide thickness.

Most Common Use of Varactors in RFIC Design:

The most prevalent application is *tuning resonant circuits*, especially Voltage Controlled Oscillators (VCOs), phase-locked loops (PLLs), and tunable filters. Varactors allow dynamic adjustment of circuit resonance, enabling frequency agility.

Evaluation of Each Type for RFIC Tuning Applications:

1. P-N Junction Varactor:

- *Pros:* High Q provides low phase noise in VCOs; Good for high-frequency applications.
- *Cons:* Not usually available in CMOS; Limited tuning range; Integration constraints.

2. MOS Varactor:

- *Pros:* Easily integrated; Wide tuning range; Useful in digital/RF co-design.
- *Cons:* Lower Q can degrade oscillator phase noise; May introduce nonlinearity and higher losses.

3. Accumulation Varactor:

- *Pros: Improved Q and linearity; Efficient use in VCOs and tunable filters; Better balance between integration and performance.*
- *Cons: Still less Q than P-N, but sufficient for many RF CMOS needs.*

Conclusion:

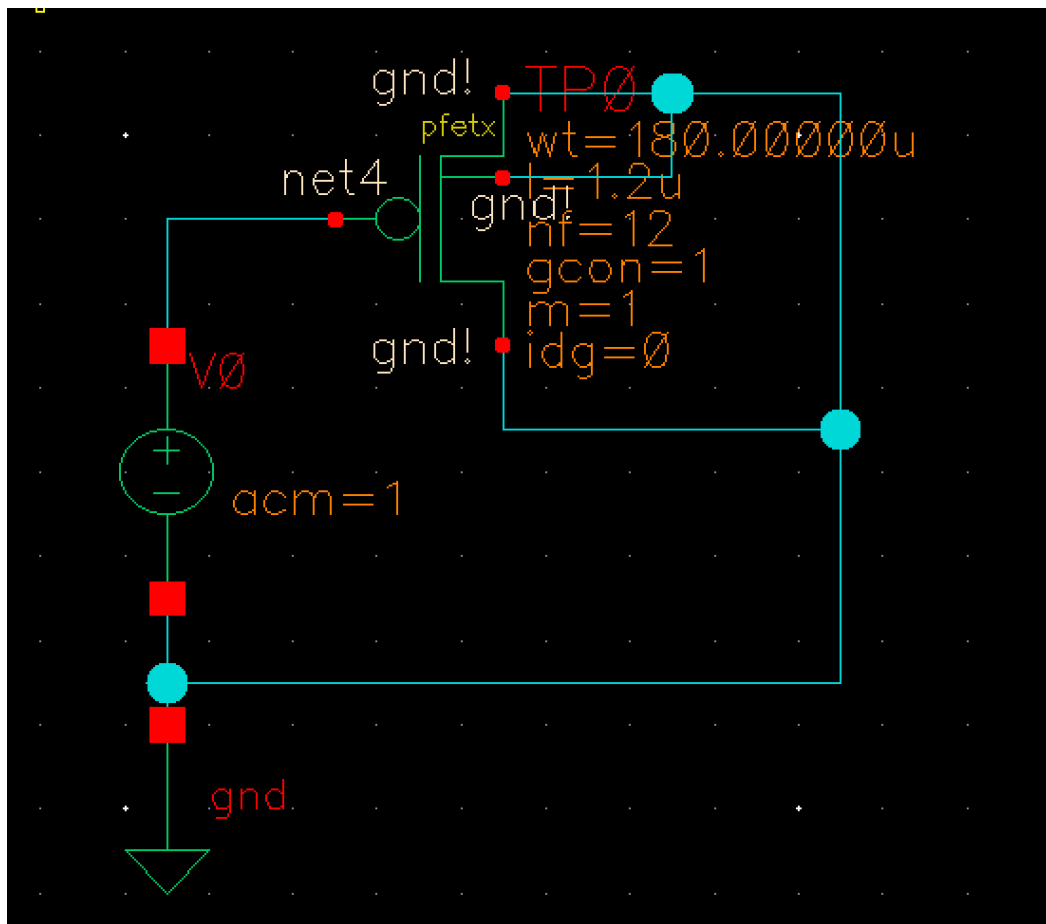
In CMOS-based RFICs, *MOS and accumulation varactors are most commonly used*, largely due to their compatibility with standard CMOS processes and sufficient performance for most VCOs and tunable circuits. While P-N junction varactors offer superior Q, their integration challenges limit their use. Between MOS and accumulation varactors, the accumulation type often strikes the best balance between tuning range, linearity, Q, and ease of fabrication for VCO tuning.

Summary Table:

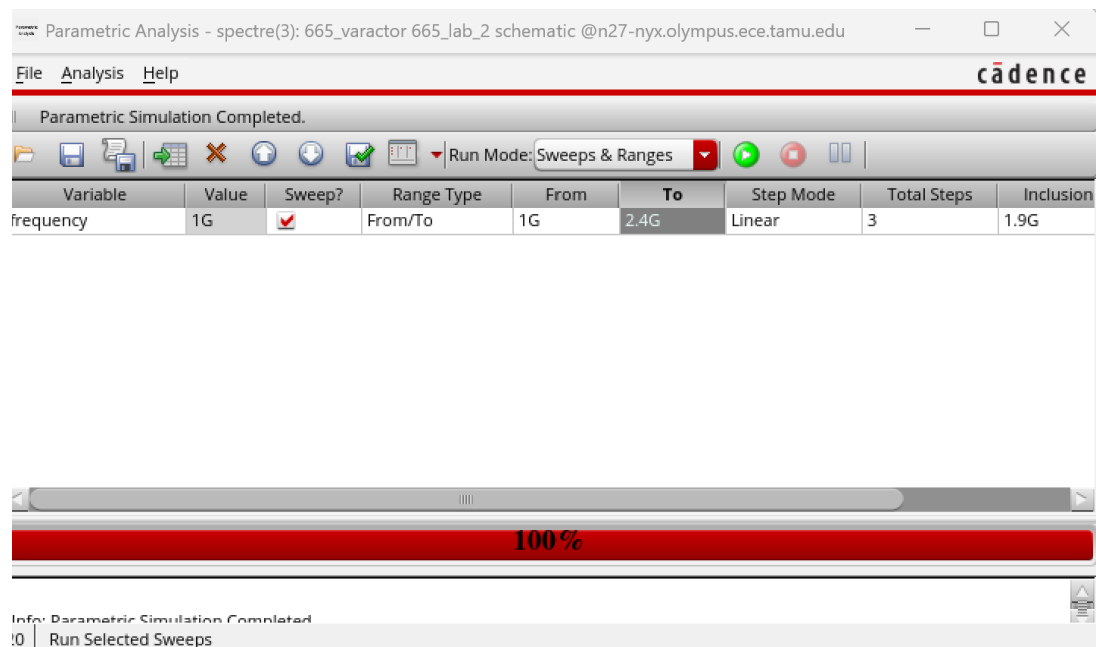
Type	Pros	Cons	RFIC Use Case Performance
P-N Junction	High Q, low phase noise	Hard to integrate, limited	Good if available, rare in CMOS
MOS	Easy CMOS integration, range	Lower Q, nonlinear	Widely used, tradeoff in noise
Accumulation	Better Q/linearity, CMOS	Still not as high-Q	Preferred in modern RFIC design

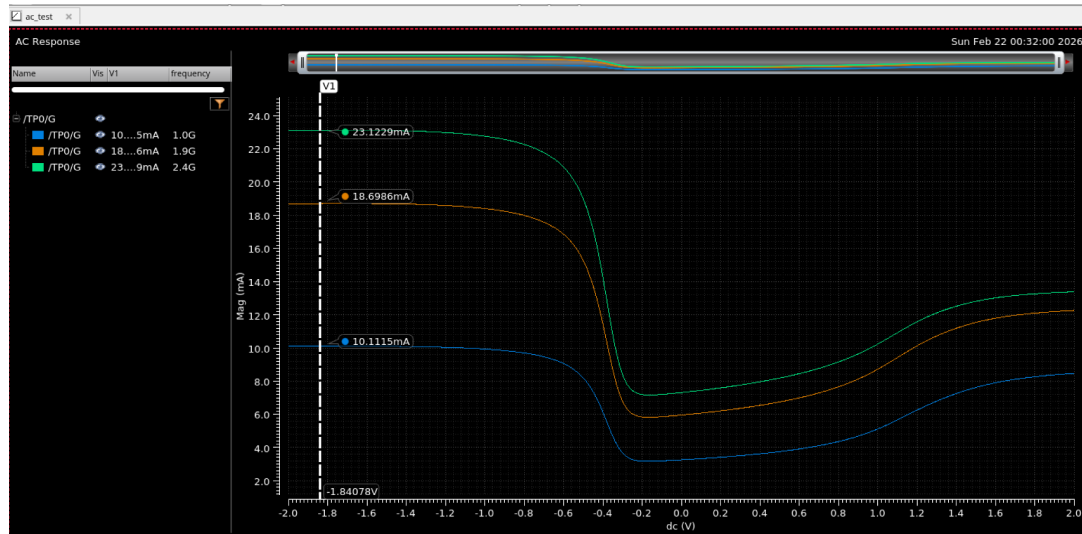
1.2 Use a PMOS transistor ($W = 12 \times 15 \mu\text{m}$, $L = 1.2 \mu\text{m}$) to create a varactor. What type of varactor is this? Obtain the Capacitance vs. V_{gs} plots of this varactor at 1GHz, 1.9GHz and 2.4GHz. Show your simulation setup and include a detailed explanation of how you obtained these characterization plots. (*Hint : Use the "family" and "value" functions of Cadence*)

Cadence Schematic Setup:



AC Analysis Setup:
Running Parametric Analysis:





2. INDUCTORS

2.1. ASITIC Installation

ASITIC is a widely used software employed for the design and optimization of integrated inductors.

The installation files and documentation can be found in:

Main page: <http://rfic.eecs.berkeley.edu/%7Eniknejad/asitic.html>

FAQ: <http://rfic.eecs.berkeley.edu/%7Eniknejad/doc-05-28-01/faq.html>

Copy the files provided by the TA in a directory named ASITIC in your UNIX account.

The file asitic_sun is the executable file. Make sure that the file is in executable mode after you transfer it to your account. If this is not the case type: `chmod 750 asitic_sun`.

The files tsmc18.tek and tsmc35.tek contain the important technology and substrate parameters required for the inductor design in tsmc 0.18um and tsmc 0.35 process. In this lab we will be using the technology file tsmc35.tek (see APPENDIX A). A sample session of ASITIC is given in APPENDIX B and a brief manual is given in APPENDIX C.

2.2 Explain the impact of inductor design parameters (metal layer, radius, metal width, etc.) on the inductor performance (inductance, quality factor, parasitics, etc.). Discuss the non-idealities and loss mechanisms in a monolithic inductor. Discuss why in new RF CMOS processes there is a dedicated metal layer for inductors? What are the main distinguishing characteristics of this metal layer?

Impact of Inductor Design Parameters:

Metal Layer:

The choice of metal layer greatly affects inductor performance. Top metal layers are thicker and further from the silicon substrate, providing lower resistance, higher quality factor (Q), and reduced capacitive coupling (parasitics) to the substrate. Thicker metals also carry more current, further reducing resistive losses.

Radius (Outer Diameter):

A larger radius increases inductance because the magnetic field encloses a larger area. However, this also increases parasitic capacitance to the substrate and between turns, which can reduce the self-resonant frequency and take up more chip area.

Metal Width:

Wider metal traces decrease series resistance, improving the Q factor, but increase parasitic capacitance and require more area. Too wide a trace may also reduce the number of turns that are possible for a given area, which can limit inductance.

Spacing Between Turns:

Larger spacing reduces parasitic (inter-turn) capacitance and eddy current losses, raising the self-resonant frequency and sometimes improving Q. However, larger spacing means fewer turns can be fitted, thus potentially reducing inductance.

Number of Turns:

More turns generally increase inductance, but add to series resistance and parasitic capacitance, which can degrade Q and lower the self-resonant frequency.

Non-Idealities and Loss Mechanisms in a Monolithic Inductor:

Series Resistance:

Due to thin, narrow metal traces and the resistivity of on-chip metals, series resistance in monolithic (on-chip) inductors lowers quality factor and increases losses.

Substrate Losses:

Capacitive coupling between the inductor and the silicon substrate causes part of the current to flow into the lossy substrate, reducing Q.

Parasitic Capacitance:

Capacitance between turns and to the substrate creates a parallel resonant path. This limits the operating frequency (self-resonant frequency) and degrades inductor performance at high frequencies.

Eddy Currents:

Magnetic fields from the inductor can induce circular currents in nearby conductive layers or the substrate, causing energy loss.

Skin Effect:

At high frequencies, current crowds to the surface of the conductor, increasing effective resistance and losses.

Dedicated Metal Layer for Inductors in New RF CMOS Processes:

Purpose:

Advanced RF CMOS processes often provide a dedicated thick top metal layer specifically for passive devices like inductors. This enhances their performance without impacting the wiring for digital circuits.

Main Distinguishing Characteristics:

Increased Thickness: The layer is thicker than standard interconnects, lowering the series resistance.

Larger Spacing from Substrate: Being the topmost layer, it increases the distance to the substrate, reducing parasitic capacitance and substrate losses.

Wider Metal Tracks: Permits the use of wider inductors for lower resistance and higher current capacity.

Special Materials: May use low-resistivity metals (like copper instead of aluminum) for further reduced resistance.

Summary:

In summary, inductor performance depends greatly on layout parameters such as metal layer, radius, width, spacing, and number of turns. Losses in monolithic inductors are mainly due to series resistance, substrate coupling, and parasitics. To address these issues, modern RF CMOS processes include a thick, top-layer metal just for inductors and other passives, characterized by its thickness, distance from the substrate, and sometimes improved materials, enabling higher Q and better performance for RF applications

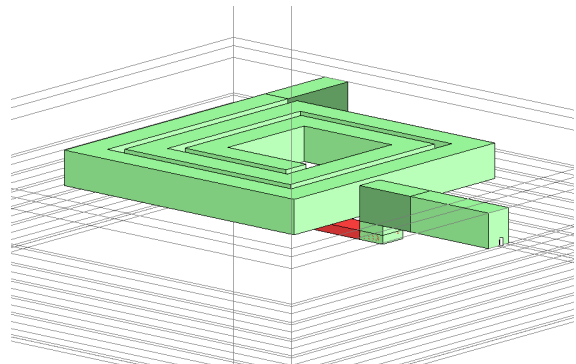
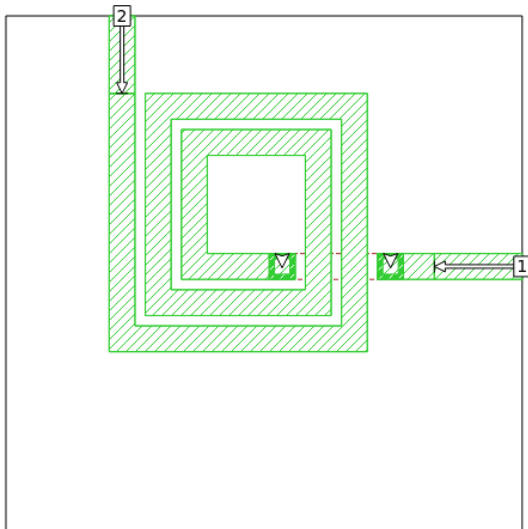
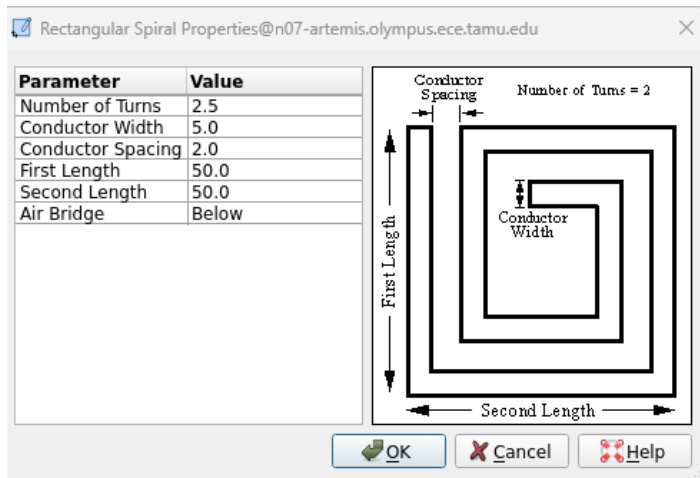
2.3 Following the sample session, design 10 different inductors (with different shapes, sizes and dimensions for the metal lines). The aim of this exercise is to see the realistic inductor values in this technology (i.e. maximum Q you can achieve with realistic inductance values, etc.) and have an idea of the area it will cost you.

Make a table to show the design parameters you picked for each inductor, include the inductance and quality factor at 1.9GHz (use "pix" command to monitor this). Discuss your observations.

(Some sample dimensions can be taken from: S. S. Mohan, et. Al., "Simple Accurate Expressions for Planar Spiral Inductances", IEEE Journal of Solid-State, Vol. 34, No. 10, October 1999.)

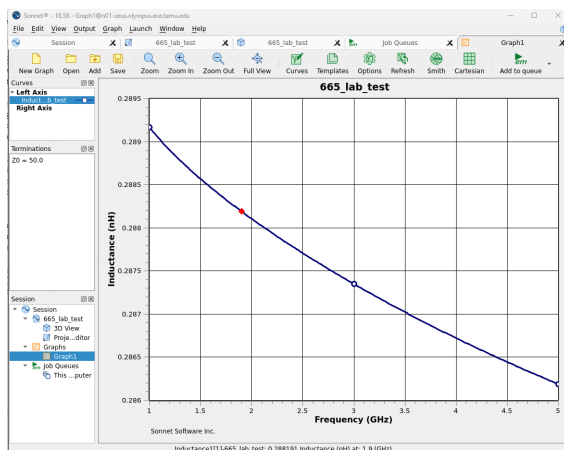
number	Inductance	Q-factor
Inductor 1	0.288	3.86533
Inductor 2	-2.69182	2.98366
Inductor 3	-0.2293	0.453349
Inductor 4	-1.28951	0.269633
Inductor 5	1.86063	4.07656
Inductor 6	0.260367	4.62978
Inductor 7	-2.33127	3.21355
Inductor 8	1.10113	8.04262
Inductor 9	-1.43751	8.63214
Inductor 10	1.37048	1.41305

Inductor design1:

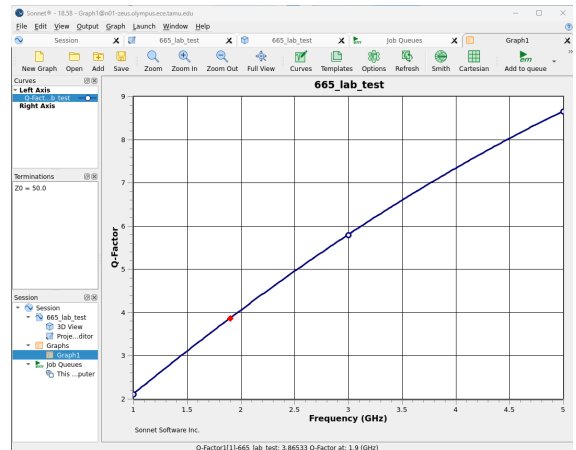


Simulation:

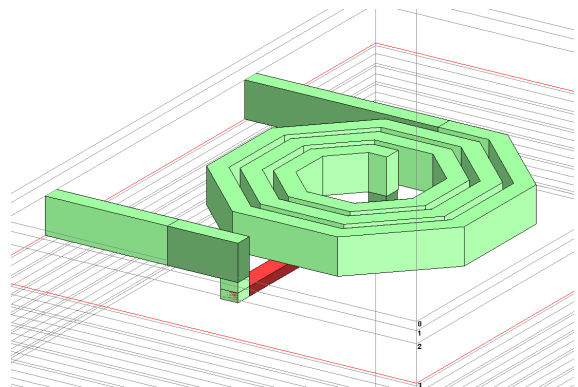
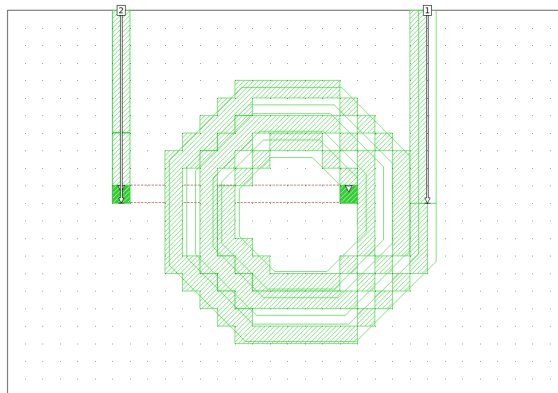
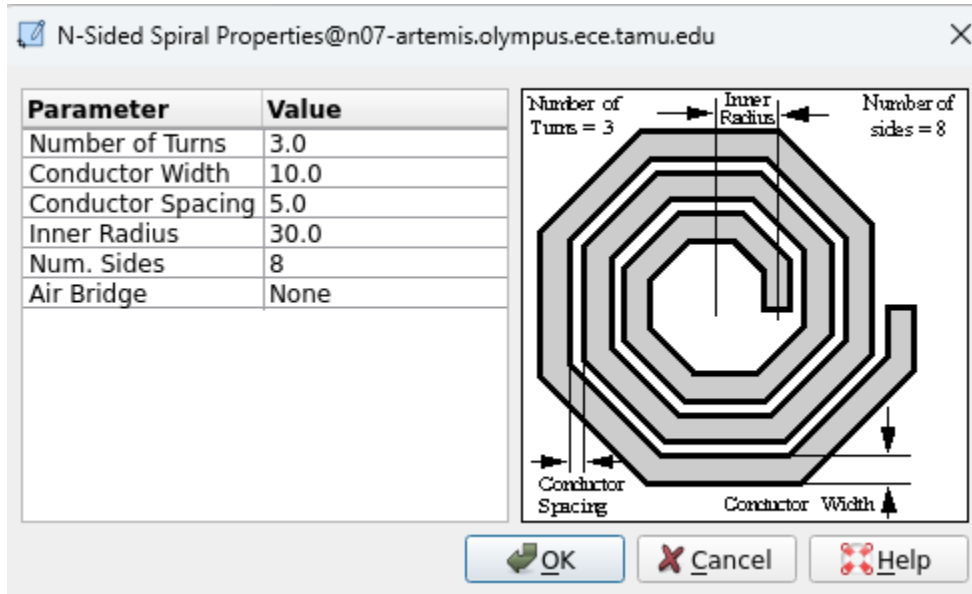
Inductance:



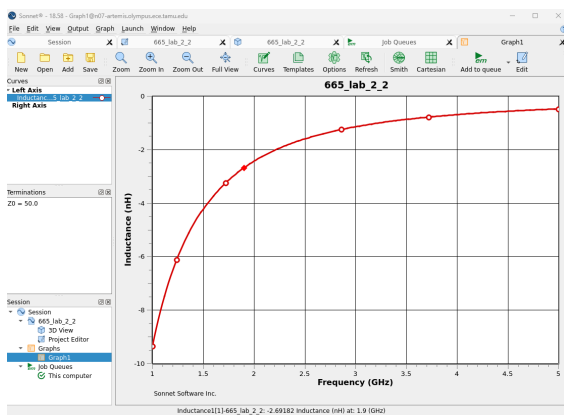
Q-factor:



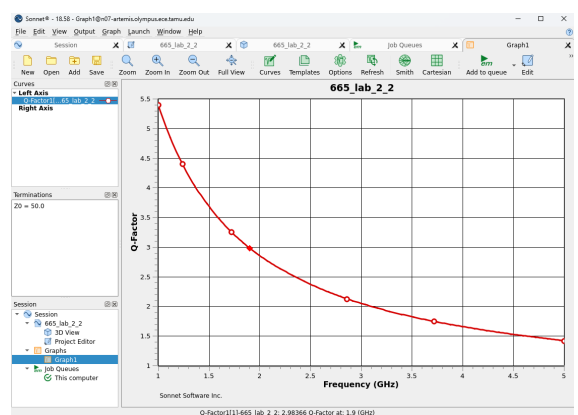
Inductor Design 2:



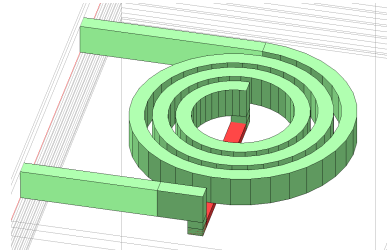
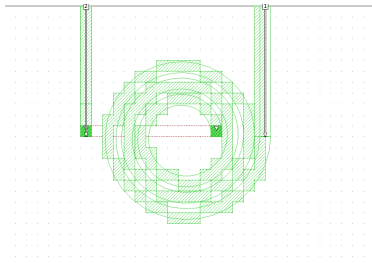
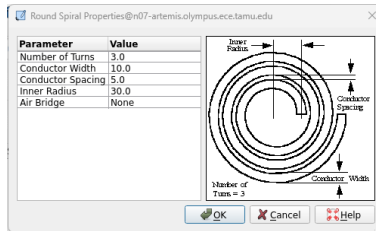
Inductance:



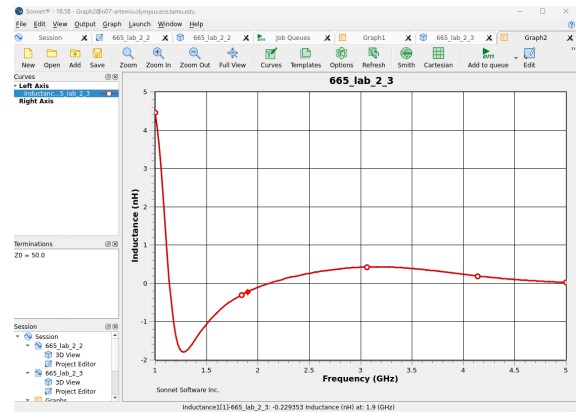
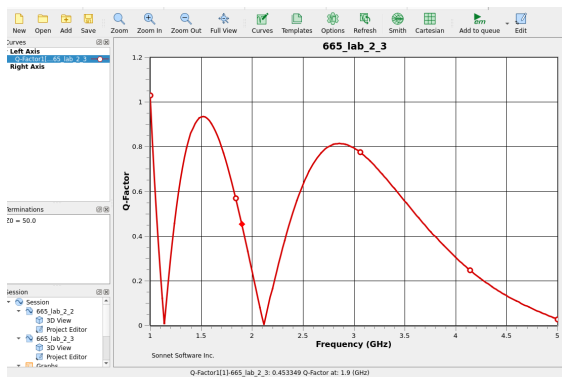
Q-factor:



Inductor Design 3:

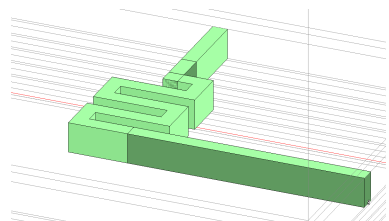
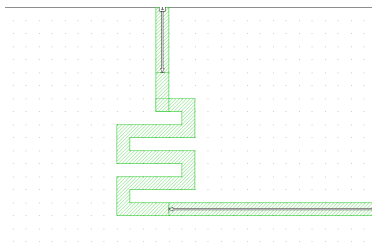
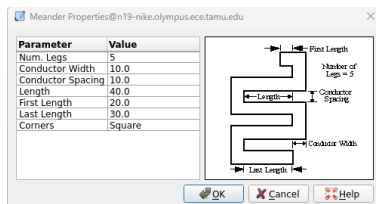


Simulation:



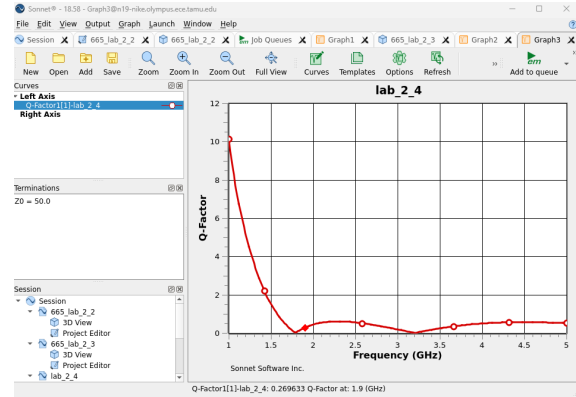
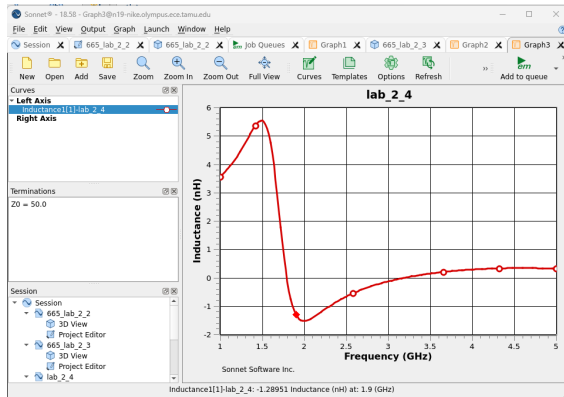
Inductor design:4

Layout:

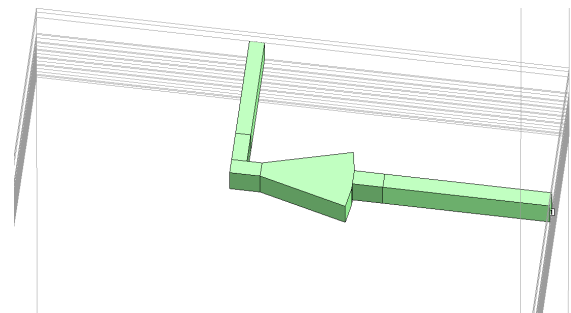
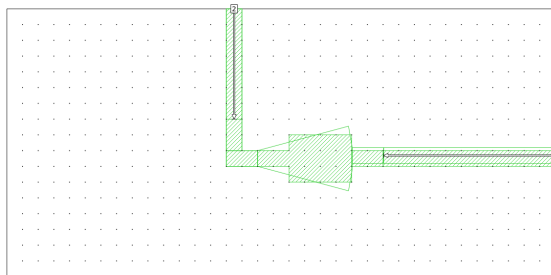
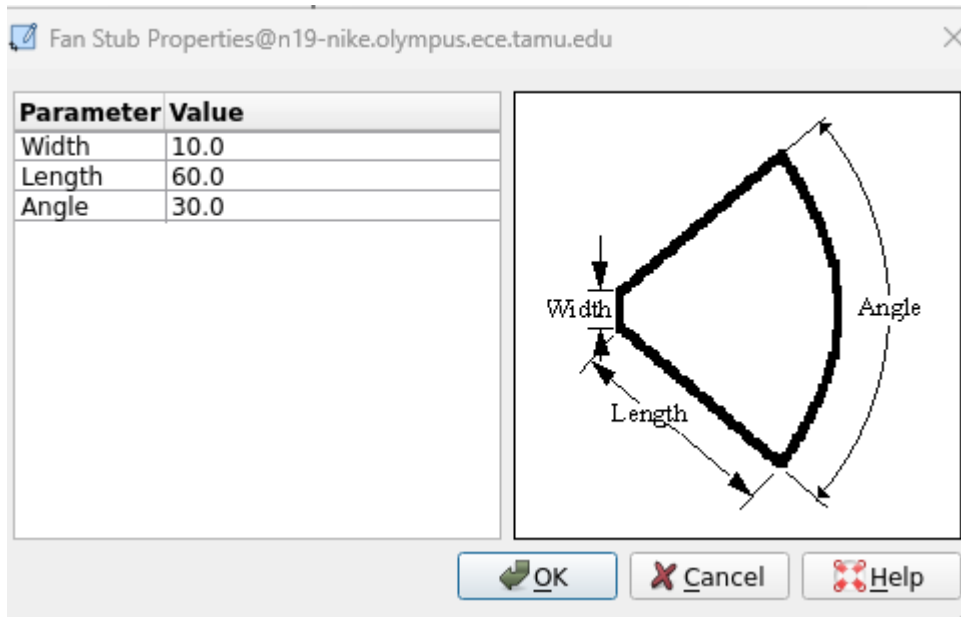


Inductance:

Q-factor:

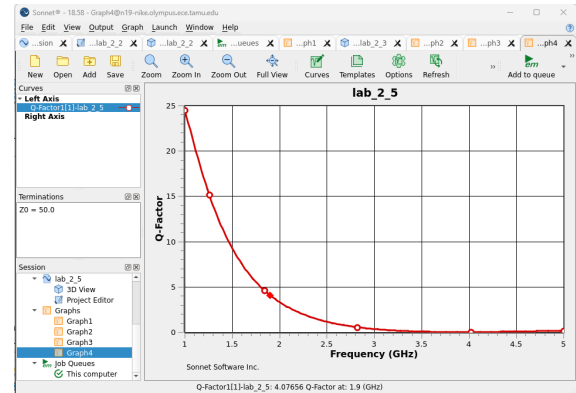
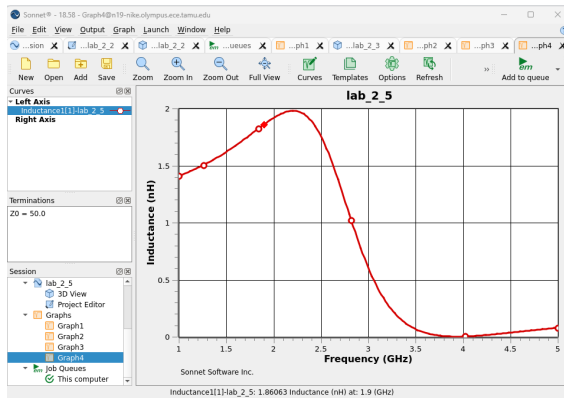


Inductor Design 5:

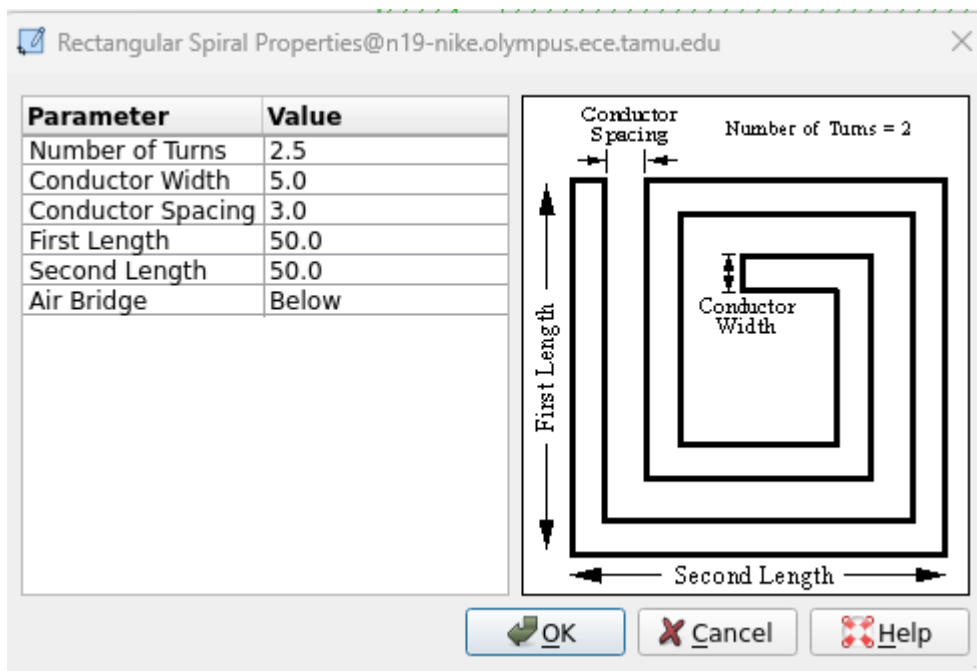


Simulation:
Inductance:

Q-factor:

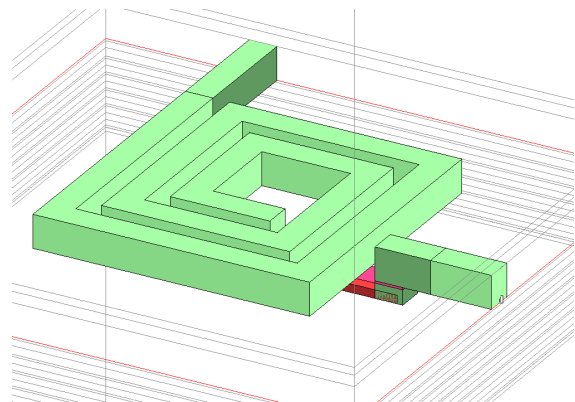
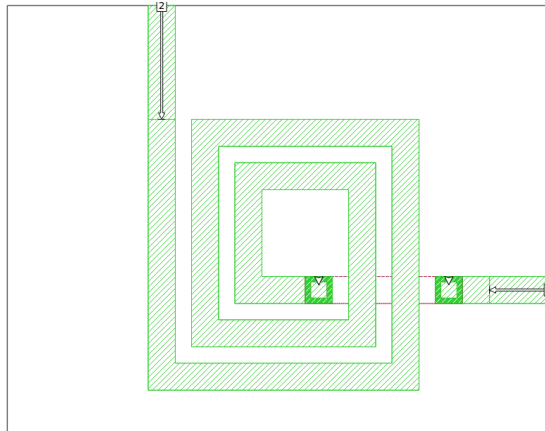


Inductor Design:6:
Increasing the spacing to 1 unit

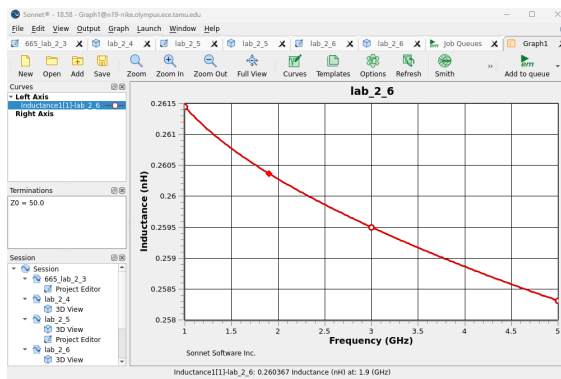


Layout:

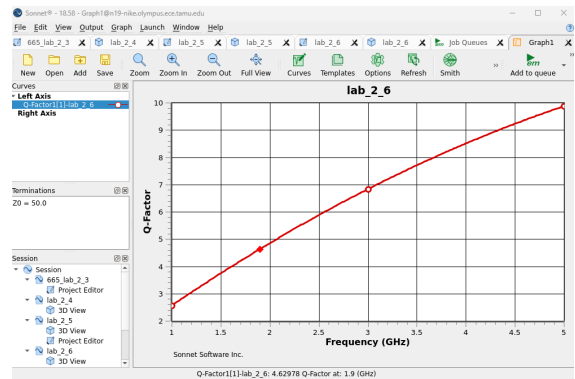
3d view:



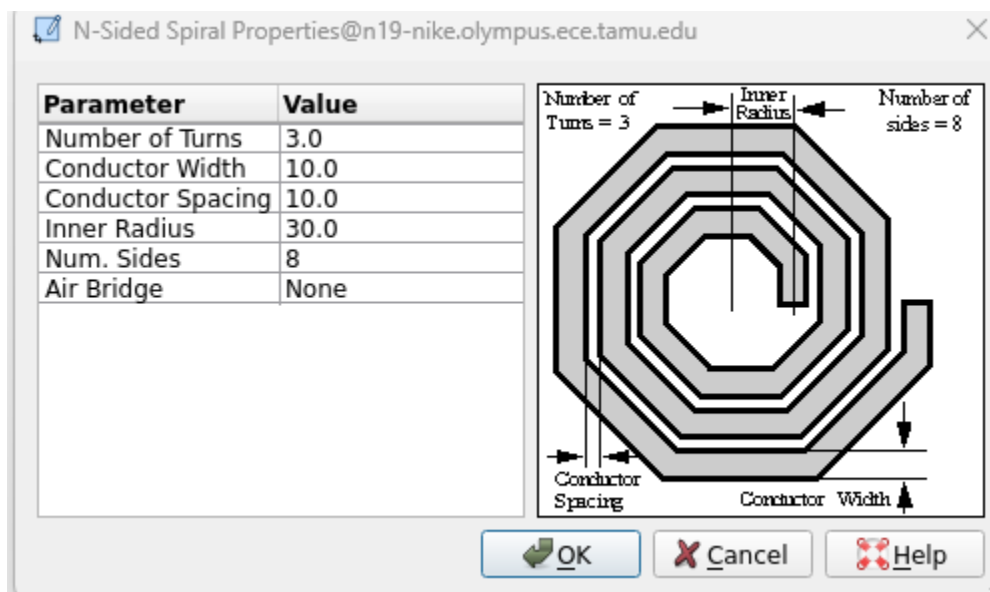
Simulation:
Inductance



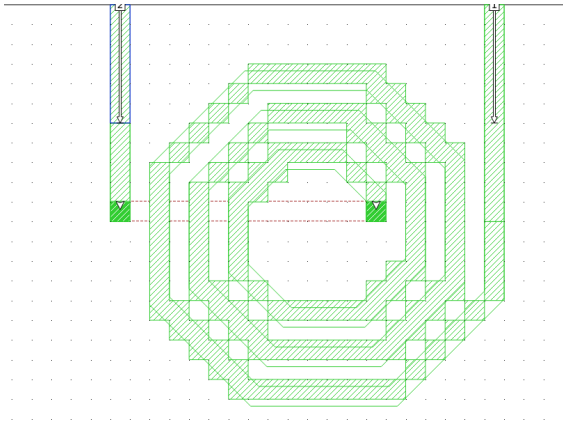
Q-Factor:



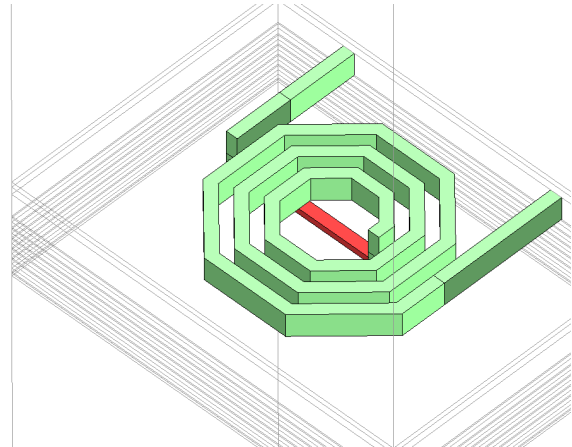
Inductor Design:7
Increasing the spacing by 5 units:



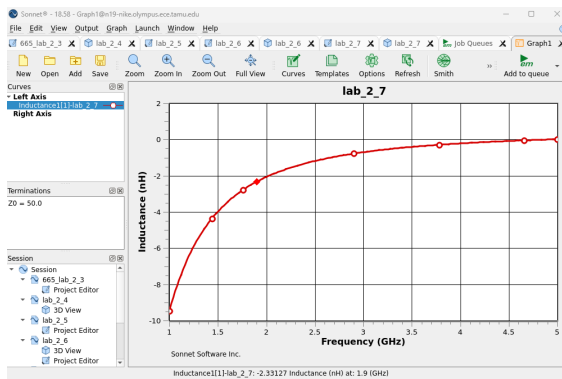
Layout:



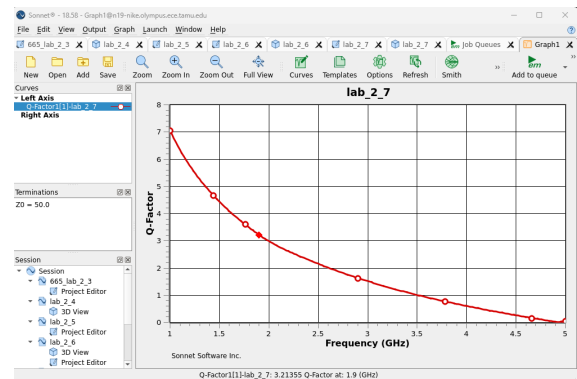
3d view:



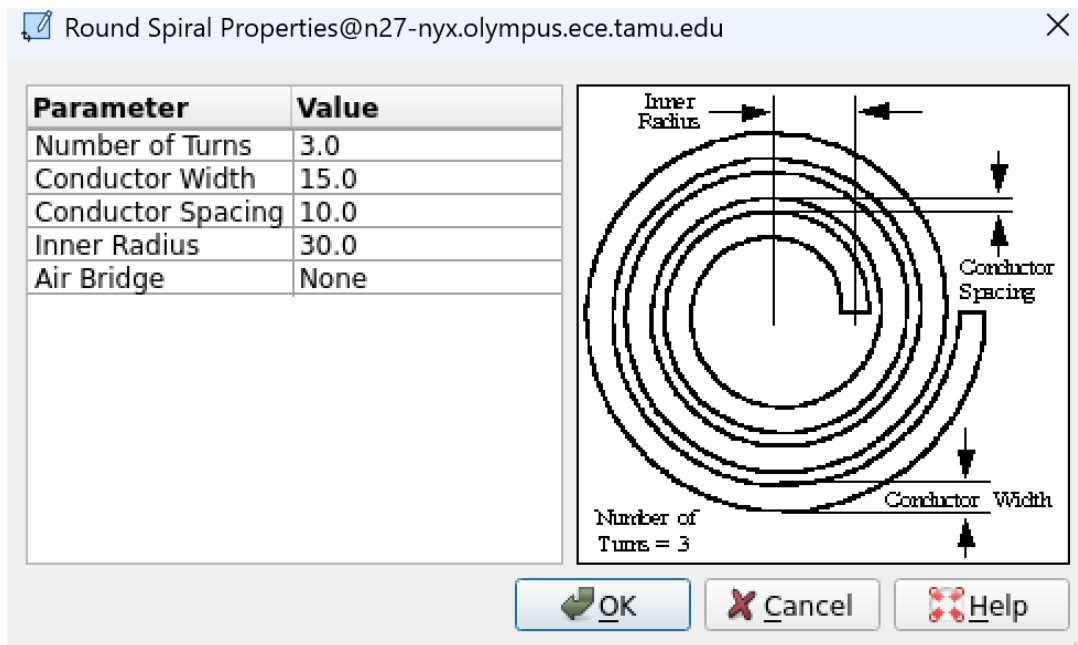
Simulations:
Inductance



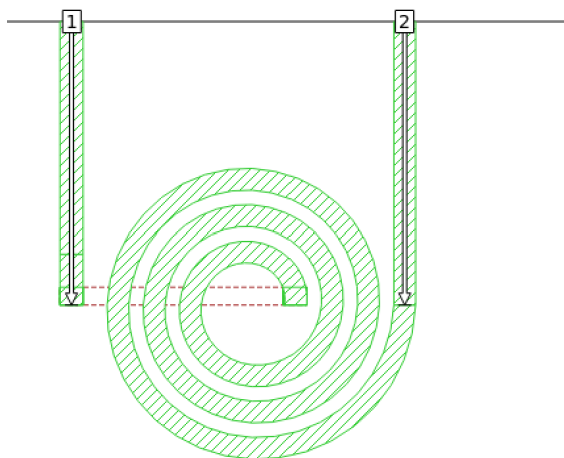
Q-factor:



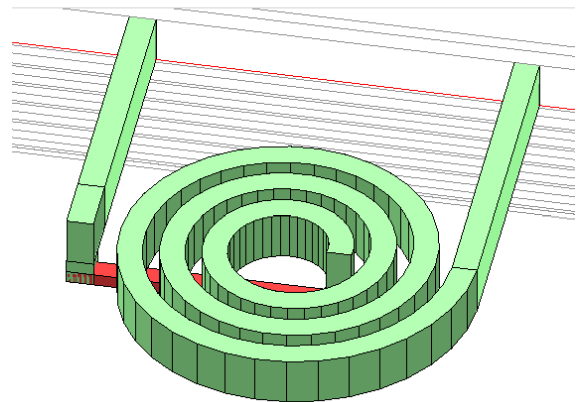
Inductor Design:8



Layout

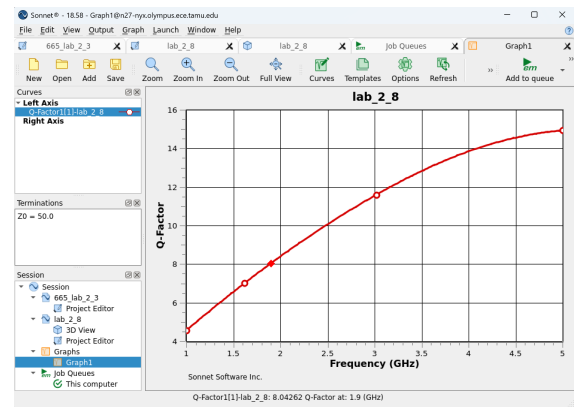
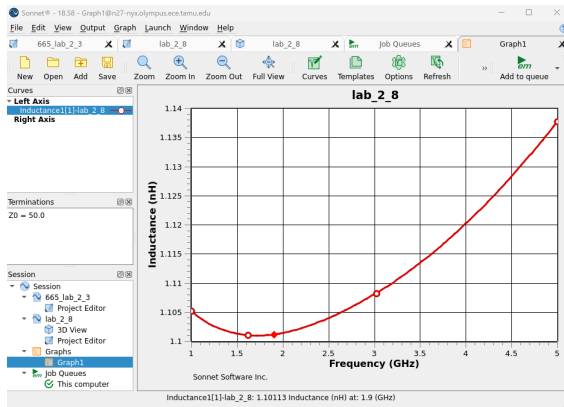


3d view:

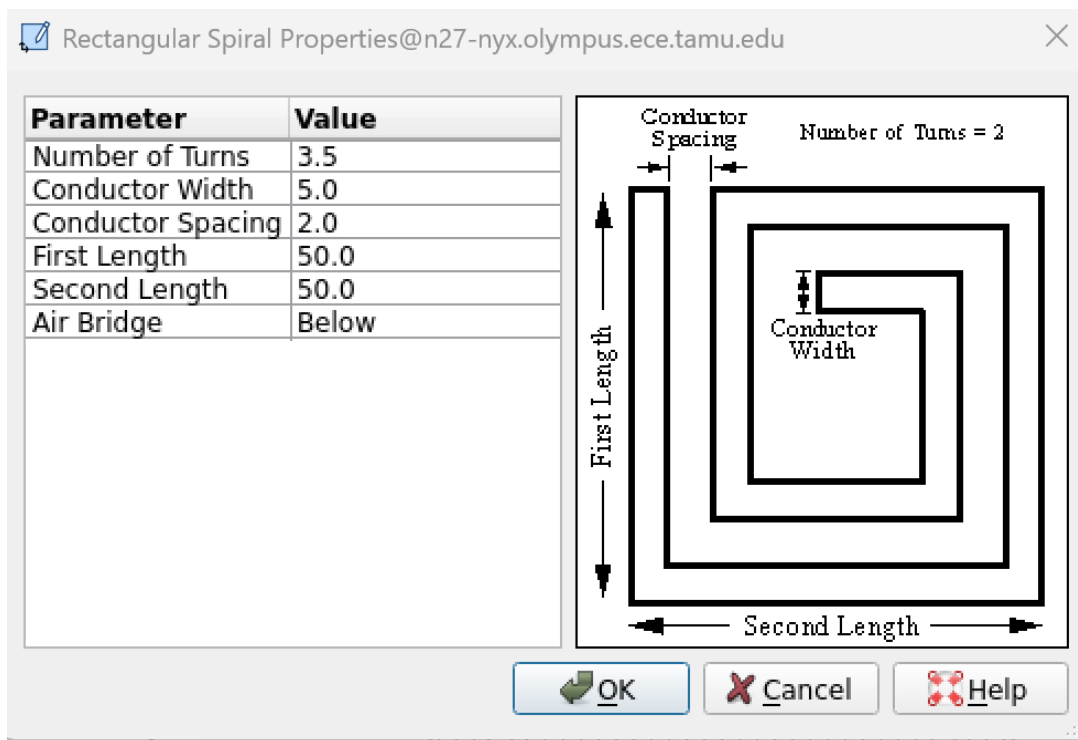


Simulation Results:
Inductance

Q-factor:

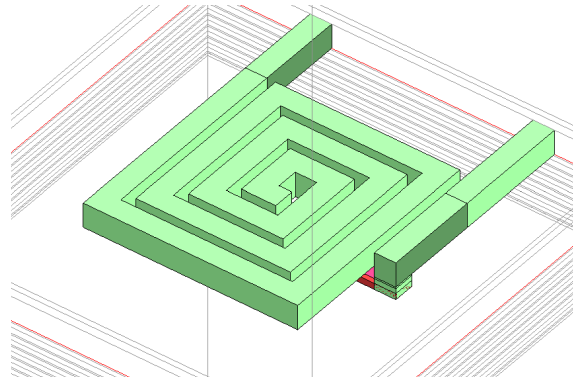
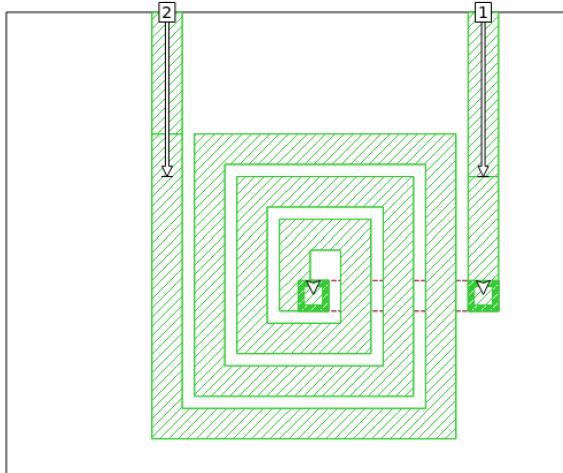


Inductor Design 9: increasing the number of turns to 3.5:

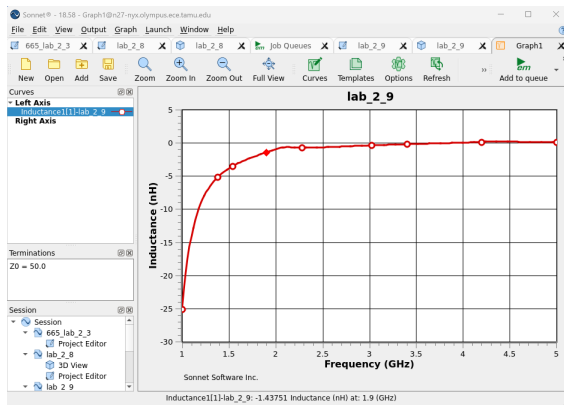


Layout

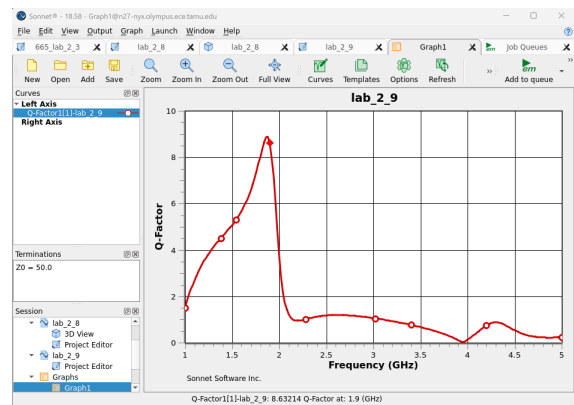
3d view:



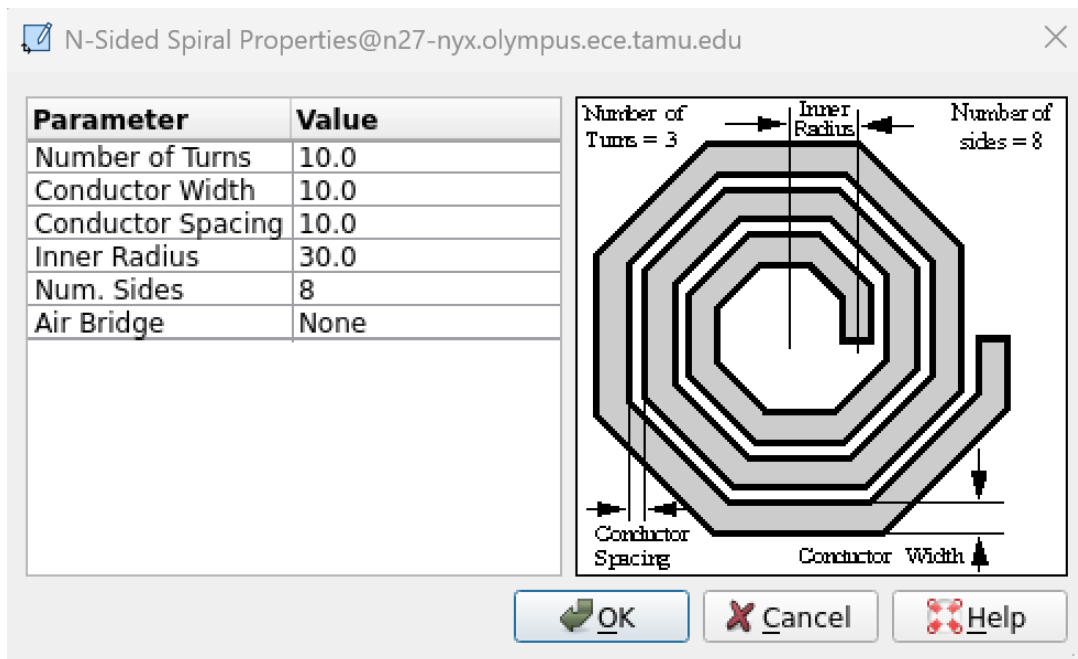
Simulation:
Inductance



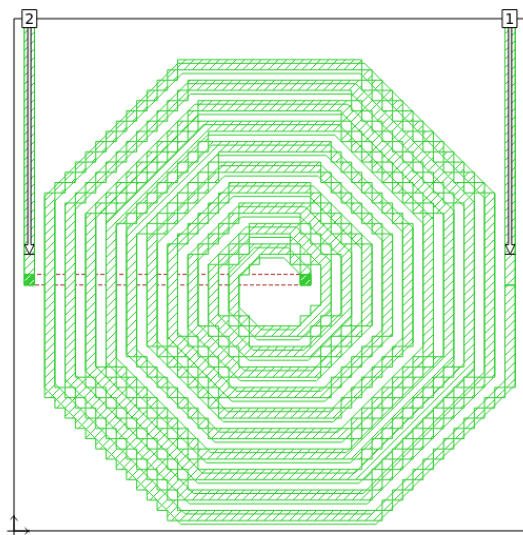
Q-Factor



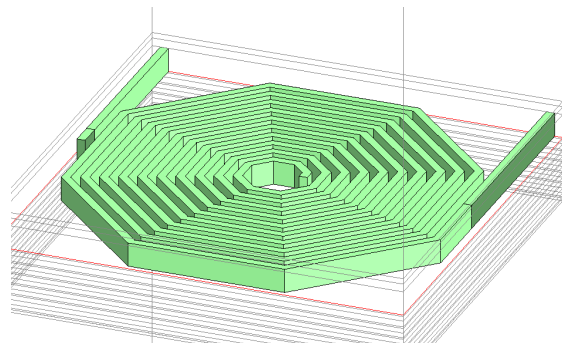
Inductor Design:10



Layout:

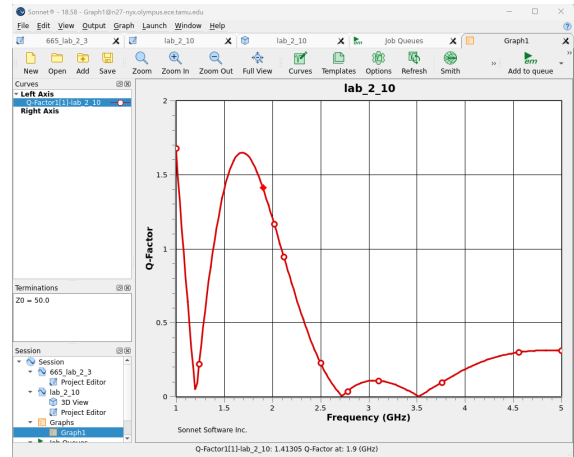
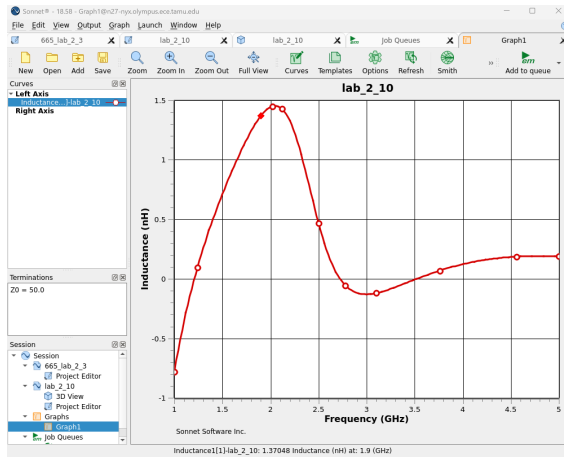


3d view:



Simulation:
Inductance

Q-factor:



3. INDUCTOR QUALITY FACTOR ANALYSIS

The quality factor Q , of an inductor is characterized by the below equation:

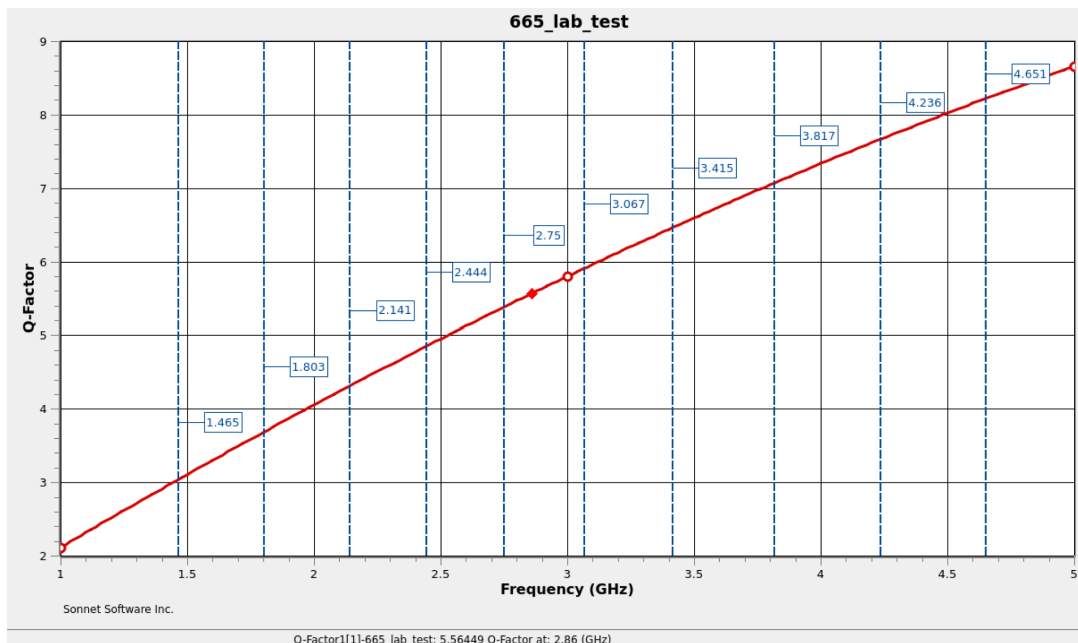
$$Q = IM(Z) / RE(Z)$$

where Z represents the impedance of the inductor and its parasitic elements.

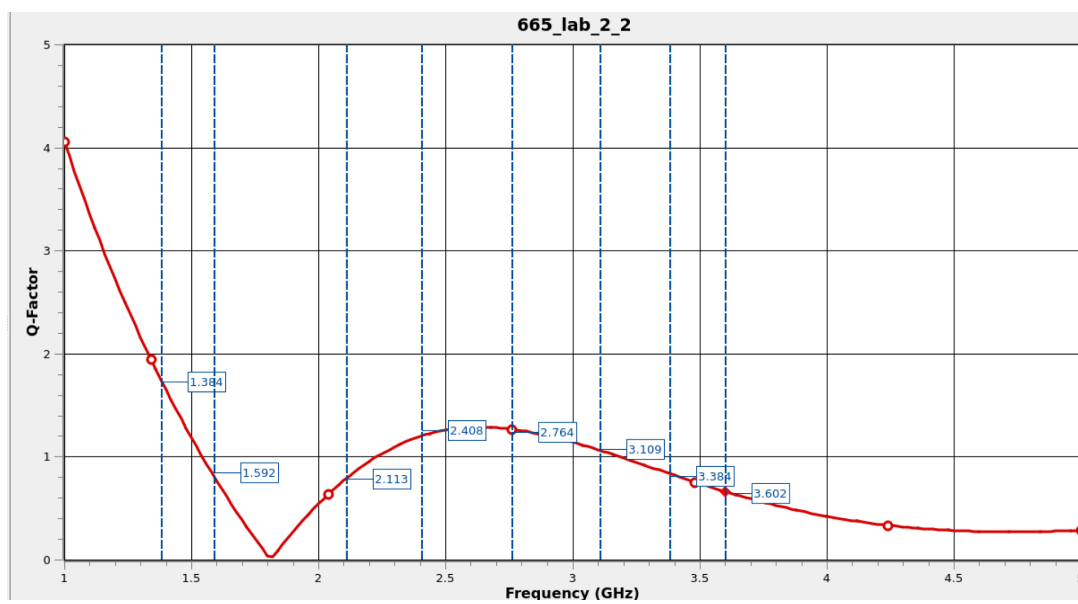
Quality Factor vs. Frequency

3.1 Pick three of the inductors you designed in section 2.3 and check the pi model and quality factor of the inductor at several different frequencies using the *pix* command to obtain a plot of Q vs. Frequency. (Note that for a given geometry, inductance will slightly change at different frequencies, as long as this change is small, you can ignore it.). What is the qualitative explanation behind the shape of the Q vs. f curve, what effects are dominant in different parts of this plot?

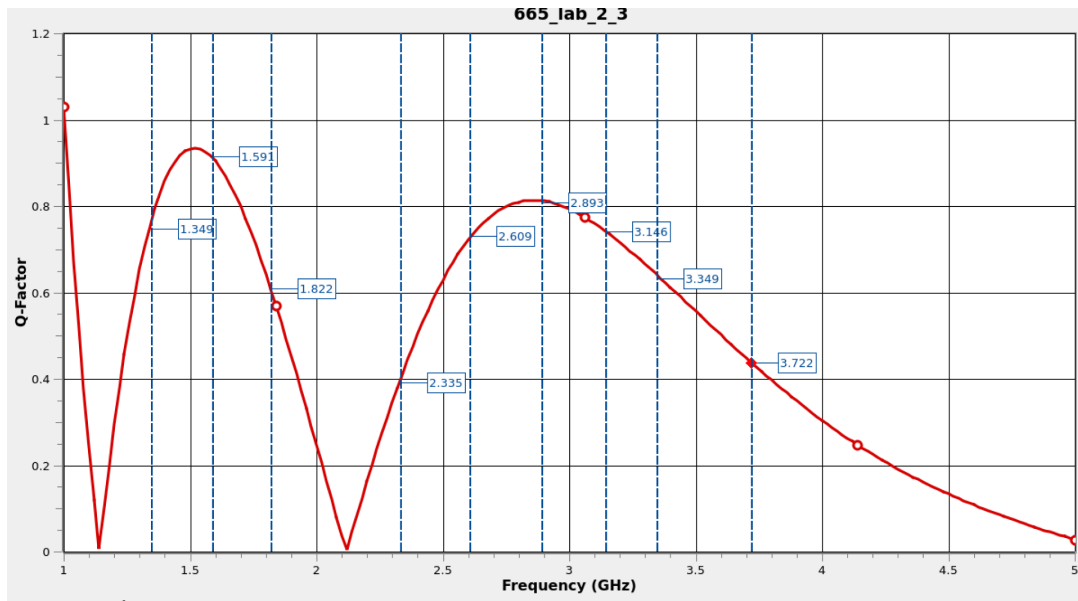
Inductor:1:Q vs Frequency plot:



Inductor_2:



Inductor 3:



The behavior of the Q factor with respect to the inductance value is of particular interest in RF design. In this section we will obtain Q vs. L curves by changing the geometric properties of inductors.

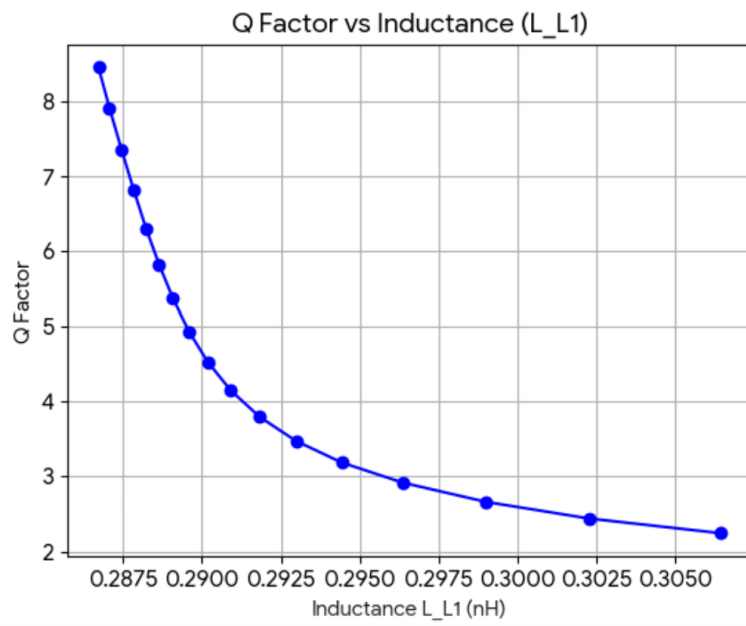
Figure 1 shows Q vs.L plot for an inductor at 900MHz. The plot is obtained by sweeping the inner radius of the inductor. The two curves also show the effect of different NT values. Figure 2 shows a sample of the inductor layout as R is swept.

3.2 Obtain Q vs. L plots as shown in the figures. Pick a sweep parameter to obtain a plot similar to Figure 1. Perform the pi model analysis for your inductors at three different frequencies to obtain three plots.

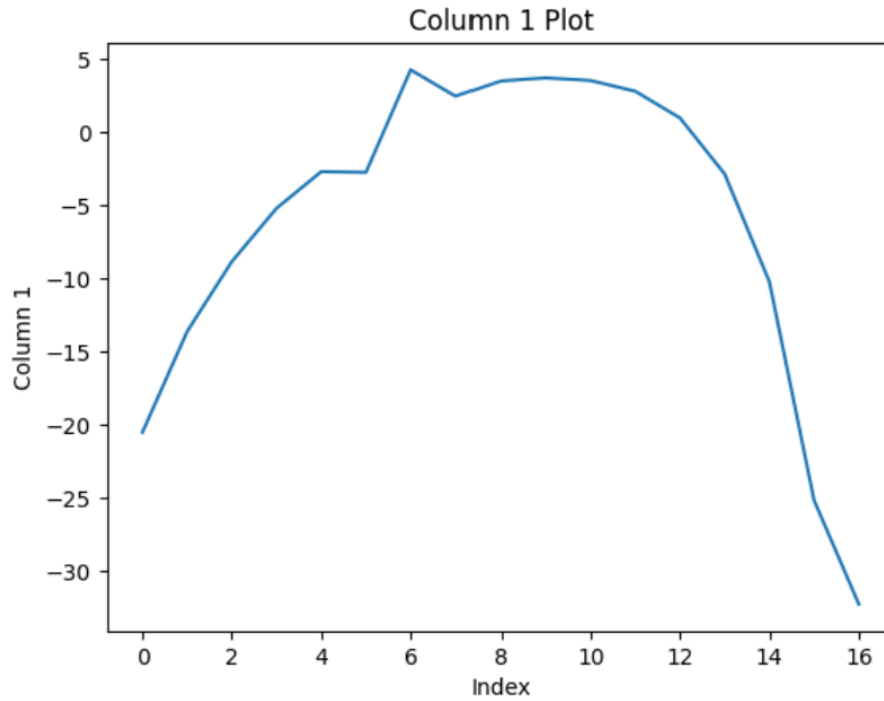
Hint : Use pix command for accuracy and keep eddy calculation mode ON for realistic Q values. Don't forget to check your resonance frequency during the pi model analysis of each inductor. The pi model calculation is not accurate and doesn't make sense unless $f_{\text{res}} \gg f_o$. So make sure all of your designs satisfy $f_{\text{res}} \geq 3f_o$

Quality Factor vs. Inductance

1.Inductor _1:



2.Inductor _2:



Inductor _3:

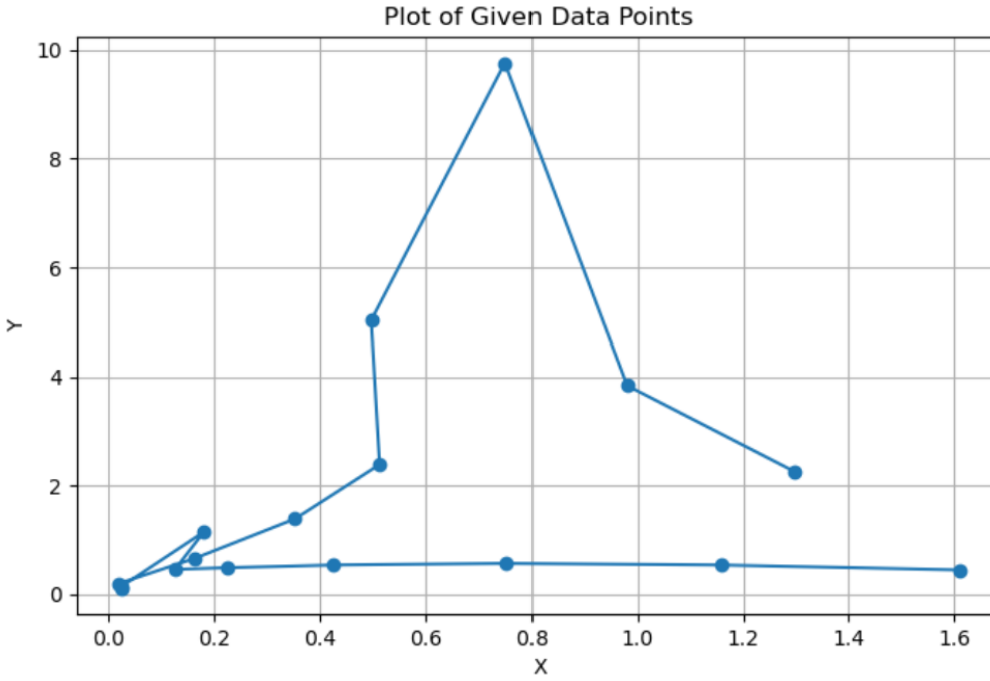


Table 1

Fig. 1 Q vs. L at 900MHz

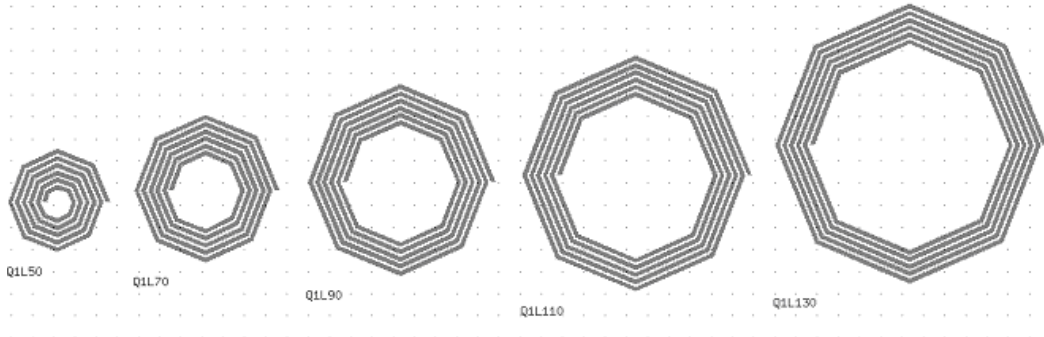


Fig 2. Q1 Series, Layouts

Inductor Quality Factor Optimization

As we saw from section 3.2, for certain frequencies or inductor ranges, Q decreases with increasing inductance value.

Let's take an inductor $L1$ with $Q1 = \omega L1 / R1$. Theoretically, if we add two of these inductors in series we will have equivalent inductance $Leq = 2L1$ but the effective Q will be

$Q_{eq} = Q1$. Figure 3. shows this.

However, we have seen that under certain conditions Q decreases for increasing inductance for a single inductor. Then, we could form large inductors from series combination of smaller inductors and therefore obtain a better Q than we would have by using one large inductor. Figure 4 represents this statement.

Table 2

Figure 3. Equivalent Q of Series Connection of inductors

Figure 4. Q improvement of a large inductor through distributed implementation

3.3 Is the above statement true? Please clearly state your answer. Support your answer with theoretical physical reasoning and ASITIC examples. You can use "join" command to perform series addition of inductors. (For any inductor you use in your explanation, give the values of all the geometry parameters for that inductor as well as the pi model results for a particular frequency. Make sure that you are doing a fair comparison between inductors.)

Below are some ASITIC tips to help with this section of the exercise. Please refer to the Appendix for detailed information on ASITIC.

- You can change the chip size that you are working on, to fit more inductors on it. Use the command "chip"
- You can hide the inductors to open some empty space in your layout, use "hide"
- To move an inductor you can either use the "move" commands (check: help edit). Or, you can select an inductor with your mouse and then drag it by clicking the middle button of your mouse to move it.
- To see the current flow directions in an inductor, use "Showdir".
- You can join two inductors in series with "join" command. Please read more about this from the ASITIC website and help documentation. Check the latest version of this documentation from the website.
- You can split joined inductors using "split" command. You can choose the segment to split the inductor from. For example, to join two inductors and then split them back to two separate inductors:

Yes, under certain conditions, this statement is true. Using a series connection of two or more smaller inductors can result in an equivalent larger inductance with a higher quality factor (Q) than what could be achieved with a single

large inductor of the same total inductance—especially in the context of on-chip (monolithic) inductors.

Theoretical Physical Reasoning

- **Quality Factor (Q) Definition:**
For an inductor with inductance L and series resistance R :
 $Q = (2\pi f \times L) / R$
where f is frequency.
- **1. Q Decreases for Larger Inductors:**
As on-chip inductance increases by making a single spiral inductor bigger (increasing the number of turns, radius, etc.), several physical effects occur:
 - Series resistance (R) increases faster than inductance due to longer conductor length, narrower traces, and more parasitic effects.
 - Substrate losses and parasitic capacitance become more severe, lowering the effective Q .
 - Eddy currents and magnetic coupling with the lossy silicon substrate are stronger in bigger inductors.
- **2. Series Combination of Identical Inductors:**
Suppose we have two identical inductors with:
 - Inductance: $L1$
 - Series Resistance: $R1$
 - Q-Factor: $Q1 = (2\pi f \times L1) / R1$
- **When connected in series:**
 - Total Inductance: $Leq = 2 \times L1$
 - Total Resistance: $Req = 2 \times R1$
 - Q-Factor: $Qeq = (2\pi f \times Leq) / Req = (2\pi f \times 2L1) / (2R1) = (2\pi f \times L1) / R1 = Q1$
- If you compare this to a single inductor with the same Leq , the resistance R' is typically greater than $2R1$ (due to physical and parasitic effects), so:
 - $Q_single = (2\pi f \times 2L1) / R'$ and since $R' > 2R1$, $Q_single < Q1$
- Therefore, building a large inductor by combining smaller ones can retain (or exceed) the higher Q of the small inductor, compared to a monolithic large spiral.

To evaluate the claim that a series connection of smaller inductors provides better or preserved Q compared to a single large inductor, we will use the inductance and resistance values from your Sonnet pi model netlist.

Example Calculation from Pi Model Data

Parameter	Single Small Inductor (.subckt SzpkYY_0)	Single Large Inductor (Theoretical)	Series of 2 Small Inductors
Frequency	1000 MHz	1000 MHz	1000 MHz
Inductance (L)	7.95 nH (L_L1)	15.9 nH (approx. double)	15.9 nH (=2 × 7.95 nH)
Series Resistance (R)	183.8 Ω (R_R1)	> 367.6 Ω (usually higher)	367.6 Ω (=2 × 183.8 Ω)
Quality Factor (Q)	0.27	< 0.27 (due to higher R)	0.27

Q Calculation:

Quality factor for an inductor can be calculated as:

$$Q = (2\pi f \times L) / R$$

For the single small inductor at 1 GHz:

$$Q = (2\pi \times 1 \times 10^9 \times 7.95 \times 10^{-9}) / 183.8 = 0.27$$

For a series combination of two such inductors:

$$L_{eq} = 15.9 \text{ nH}, R_{eq} = 367.6 \Omega$$

$$Q_{series} = (2\pi \times 1 \times 10^9 \times 15.9 \times 10^{-9}) / 367.6 = 0.27$$

For a *single* large inductor with the same L_{eq} , the resistance R_{eq} tends to be higher than the simple sum due to increased parasitics and losses:

$$Q_{large} \approx (\text{slightly less than } 0.27)$$

Conclusion and Relevance:

- The pi model data shows that the Q of a series combination of smaller inductors is equal to that of each individual small inductor.
- In practice, a single large inductor with the same total inductance experiences more resistive and substrate losses due to its geometry, leading to a lower Q.
- Thus, series-connecting smaller inductors preserves higher Q, as shown in your simulation data, and is a valid strategy in on-chip RFIC design.

How to Use the Pi Model Data:

1. Find L and R values at your frequency of interest from the .subckt entry (e.g., "L_L1" and "R_R1").

2. Use the formula $Q = (2\pi f \times L)/R$ for each inductor.
3. Add inductors in series: $L_{eq} = L + L \dots$, $R_{eq} = R + R \dots$
4. Compare the Q of the combined inductor vs. a single large design.
5. The simulation demonstrates that splitting inductors can result in improved or preserved Q (and thus better circuit performance).

Final Remarks:

Your reports for this lab are due next week's lab session.